

Maggie (Wanlin) Tu

Data Scientist & Applied AI Engineer

maggietu570@gmail.com · linkedin.com/in/maggie-tu · github.com/WanlinTu · Nashville, TN · Open to Relocation · 513-764-2999

SUMMARY

Data Scientist with experience across NLP research, data engineering, and financial analytics. Built an end-to-end LLM pipeline for SEC filing sentiment extraction achieving a 2.02 out-of-sample Information Ratio. Background spans applied AI, large-scale language analysis, Azure data pipelines, and risk analytics. M.S. Data Science, Vanderbilt University (May 2026). Open to Data Scientist, Applied AI, and Analytics roles.

EDUCATION

M.S. Data Science

May 2026

Vanderbilt University – Data Science Institute · Nashville, TN

B.S. Business Analytics

December 2023

University of Cincinnati · Cincinnati, OH

B.S. Finance (Investment Banking), Transfer Coursework — Chengdu University of Technology

EXPERIENCE

Data Engineer Intern

May 2025 – Aug 2025

FusionSite Services · Nashville, TN

- Built Azure Function App pipelines and PostgreSQL workflows automating data ingestion across safety, HR, and training systems
- Analyzed 15,000+ driver safety events across 56 sites, identifying repeat-risk patterns for regional leadership
- Evaluated driver improvement program effectiveness across 14,000+ cleaned records and 250+ enrolled drivers
- Built executive Power BI dashboards tracking headcount, turnover, training completion, and workforce KPIs
- Integrated DigitalChalk LMS training data into reporting workflows for site-level compliance tracking

Risk Analyst Intern

March 2024 – June 2024

Protiviti · Cincinnati, OH

- Conducted AML investigations on 50+ accounts per week, analyzing customer behavior, transactions, and KYC documentation
- Produced 200+ investigative risk narratives, maintaining quality standards at 8 investigations per day
- Reviewed 10,000+ transactions for money laundering, fraud, sanctions, and compliance concerns

Data Analyst Intern

June 2023 – August 2023

WaterFire Providence · Providence, RI

- Evaluated 200+ fundraising campaigns in R and Tableau to identify top revenue generators over 3 years
- Analyzed 8,000+ donors in Python and Excel to identify lapsed donors for targeted fundraiser
- Detected data redundancies and improved fundraising campaign data infrastructure

PROJECTS

LLM-Powered Financial Signal Extraction from SEC Filings

- Engineered a corpus of 67K+ financial sentiment factors from 2,441 SEC filings (2015–2025) using DeepSeek-R1-Distill-Qwen-14B
- Fine-tuned LLM via QLoRA on 5,217 Claude Opus-annotated samples; tail F1 doubled (very_negative: 0.025→0.050)
- Applied GRPO reinforcement learning to align sentiment with real stock returns; L/S spread improved 2.78%→8.09%

- Achieved 2.02 HAC-valid Information Ratio on held-out 2023–2025 data; 11/12 OOS scenarios met or exceeded IS performance
- Pre-registered 48-cell backtest grid across 4 strategies × 4 variants × 3 horizons before observing test results

Large-Scale Mental Health NLP Research (HLAB, Vanderbilt)

- Built NLP workflows to analyze 918K+ Reddit submissions using DLATK, LDA topic modeling, and logistic regression
- Applied Differential Language Analysis to 50K+ n-gram features, identifying language markers for PTSD, cPTSD, and anxiety
- Generated 100-topic latent semantic representations; analyzed longitudinal trauma discourse trends 2020–2025

SKILLS

Languages & Data	Python, SQL, R, PySpark, Snowflake, Databricks, Azure, MySQL
AI & ML	LLMs, NLP, QLoRA Fine-Tuning, GRPO / Reinforcement Learning, DeepSeek-R1, GPT-4o, HuggingFace, PyTorch, vLLM
Analytics & BI	Power BI, Tableau, Excel, SEC EDGAR processing, DLATK, Git
Languages	English (fluent), Mandarin Chinese (native)

maggietu570@gmail.com · linkedin.com/in/maggie-tu · Nashville, TN · Open to Relocation